

Nathaniel Billig

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EDUCATION

University of Wyoming, College of Engineering and Physical Sciences

Master of Science: Electrical Engineering

Expected May 2027

Cumulative GPA 3.9/4.0

Relevant Coursework: *Advanced Robotics Modeling, Robot Learning, Robot Perception, Cooperative Robotics, AI for Multiagent Systems*

University of Wyoming, College of Engineering and Physical Sciences

Bachelor of Science in Computer Engineering; Minors in Computer Science and Mathematics

May 2025

GPA 3.8/4.0

Relevant Coursework: *Advanced Microprocessors, HDL Digital Design, Neural Networks, Deep Reinforcement Learning*

EXPERIENCE

Robotics Researcher — USSF SSTI Sponsored Research

Sept 2024 – Present

University of Wyoming

- Designed and built a three-finger, tendon-driven robotic gripper for USSF-sponsored research, including CAD modeling, sensor selection and integration, and embedded software.
- Designed a four-layer ESP32 controller PCBA in Altium CircuitMaker with motor drivers, load-cell, and Hall-effect sensor interfaces, thermal vias, and separation between 24V power and sensitive signal traces; completed schematic design, layout, and manufacturing submission.
- Integrated the gripper with a Kinova Gen3 seven-degree-of-freedom arm for static and moving-target grasping tests.
- Developed three iterations of a tendon-driven robotic gripper, revising finger geometry, sensor placement, and mechanical design based on testing and sponsor feedback.
- Evaluated grasping across seven objects with different geometries, achieving 60 successful grasps in 70 laboratory trials (85.7%).
- Developed simulation and calibration methods to characterize gripper kinematics, joint sensing, and contact forces.

Microcontroller Project Engineer Intern, 4th State Communications – Cheyenne, WY

Oct 2023 – Feb 2024

- Developed C/C++ Arduino firmware for UART communication and control of telecommunications transmitters.
- Built Python software using Linux NFS to transmit control commands between a central controller and networked relay systems.

Undergraduate Research Assistant, University of Wyoming – Laramie, WY

May 2023 – Sept 2025

- Developed computer vision applications and embedded Python/C/C++ software for Arduino and Raspberry Pi-based autonomous platforms.
- Instructed high school students and teachers in physical computing, programming fundamentals, and hands-on electronics.

PROJECTS

Hall Effect Robotic Joint Angle Sensing, University of Wyoming

- Developed Hall-effect joint-angle sensing over a 0–90° range, achieving sub-degree accuracy through calibration and manual verification; tested measurement repeatability.

Multi-Agent Continuum Arm Q-Learning, University of Wyoming

- Developed a multi-agent Q-learning framework for coordinated control of a continuum robotic arm, evaluating discrete state and action spaces for joint coordination and motion control.

Robotic Gripper Modeling, University of Wyoming

- Built a MATLAB/Simscape model of the gripper, compared simulated joint motion with experimental measurements, and estimated contact forces during grasping.

PUBLICATIONS & PRESENTATIONS

N. Billig, S. Smith, and I. D. Walker. “A Soft Adaptive Gripper for Non-Cooperative Object Grasping.” *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2026. Presented October 2026.

N. Billig and I. D. Walker. “Design of a Tendon Driven Underactuated Robotic Gripper for Acquisition of Non-Cooperative Objects.” *IEEE Aerospace Conference*, 2027. Submitted; under review.

HONORS & ACTIVITIES

Crowd Favorite Poster — USSF University Consortium Symposium, 2026

Computer Engineering Student of the Year — Wyoming Engineering Society, 2025

Tau Beta Pi Engineering Honor Society — Inducted 2024

University of Wyoming Robotics Club — End-effector team member for the University Rover Challenge.

SKILLS

Programming & Embedded Systems: C/C++, Embedded Programming, Python, MATLAB, Arduino, Raspberry Pi, ESP32

Electronics & Communication: UART, SPI, I²C, PCB/Schematic Design, Sensor Integration, Soldering

Robotics & Simulation: ROS, Kinematics, Path Planning, Digital Twins, Reinforcement Learning, Computer Vision

Software & CAD: SolidWorks, Fusion 360, CircuitMaker, OpenCV, MATLAB/Simulink, Simscape, Microsoft Office Suite